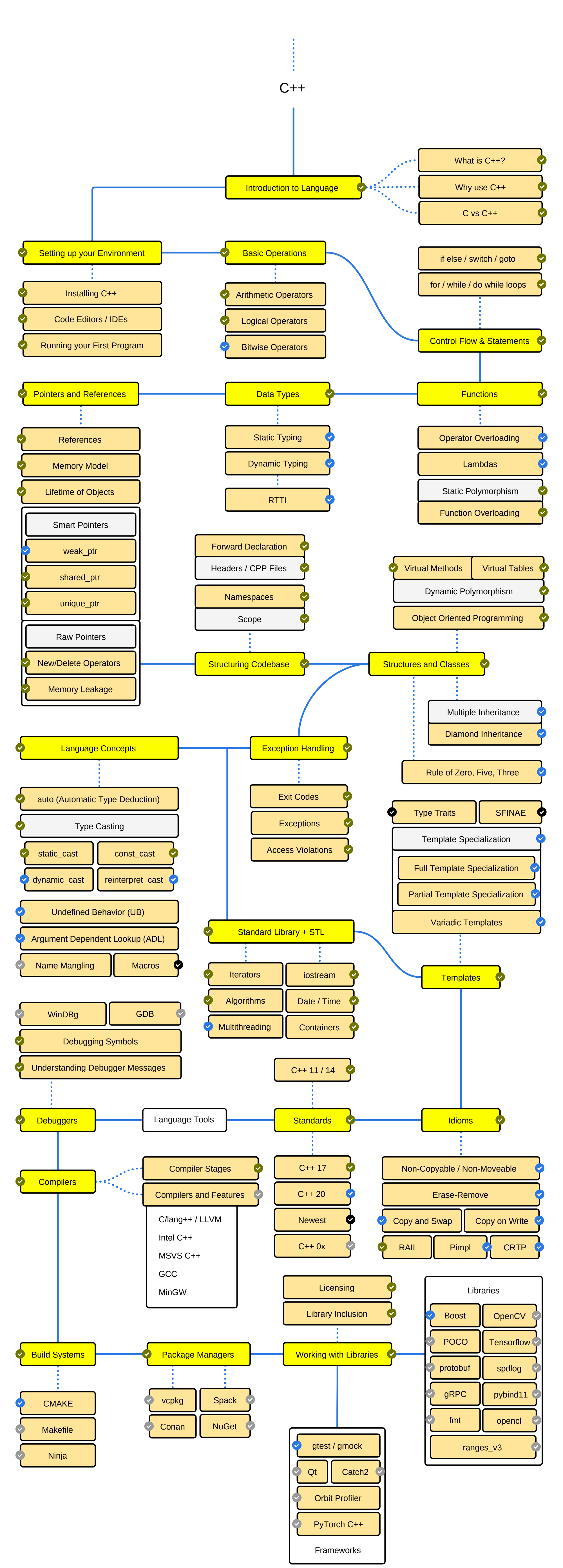
C++
Introduction to Language
What is C++?
Why use C++
C vs C++
Setting up your Environment
Installing C++
Code Editors / IDEs
Running your First Program
Basic Operations
Arithmetic Operators
Logical Operators
Bitwise Operators
if else / switch / goto
for / while / do while loops
Control Flow & Statements
Pointers and References
References
Memory Model
Lifetime of Objects
Smart Pointers
weak_ptr
shared_ptr
unique_ptr
Raw Pointers
New/Delete Operators
Memory Leakage
Data Types
Static Typing
Dynamic Typing
RTTI
Functions
Operator Overloading
Lambdas
Static Polymorphism
Function Overloading
Forward Declaration
Headers / CPP Files
Namespaces
Scope
Structuring Codebase
Virtual Methods
Virtual Tables
Dynamic Polymorphism
Object Oriented Programming
Structures and Classes
Multiple Inheritance
Diamond Inheritance
Rule of Zero, Five, Three
Language Concepts
auto (Automatic Type Deduction)
Type Casting
static_cast
const_cast
dynamic_cast
reinterpret_cast
Undefined Behavior (UB)
Argument Dependent Lookup (ADL)
Name Mangling
Macros
Exception Handling
Exit Codes
Exceptions
Access Violations
Type Traits
SFINAE
Template Specialization
Full Template Specialization
Partial Template Specialization
Variadic Templates
Standard Library + STL
Iterators
iostream
Algorithms
Date / Time
Multithreading
Containers
Templates
WinDBg
GDB
Debugging Symbols
Understanding Debugger Messages
C++ 11 / 14
Debuggers
Language Tools
Standards
Idioms
Compilers
Compiler Stages
Compilers and Features
C/lang++ / LLVM
Intel C++
MSVS C++
GCC
MinGW
C++ 17
C++ 20
Newest
C++ 0x
Non-Copyable / Non-Moveable
Erase-Remove
Copy and Swap
Copy on Write
RAII
Pimpl
CRTP
Licensing
Library Inclusion
Libraries
Boost
OpenCV
POCO
Tensorflow
protobuf
spdlog
gRPC
pybind11
fmt
opencl
ranges_v3
Build Systems
Package Managers
Working with Libraries
CMAKE
Makefile
Ninja
vcpkg
Spack
Conan
NuGet
gtest / gmock
Qt
Catch2
Orbit Profiler
PyTorch C++
Frameworks